Lab 7: Fuzzing for Input Validation Bugs

Dabbugunta Venya Anand | Python 3.12.14 | pytest 9.2.0 | Hypothesis 6.170.14

Objective

Use Hypothesis to identify input-validation crashes in the assigned processor, fix the implementation, and verify that valid inputs still produce correct results.

Before the fix

The original application crashes on **1:0** with ZeroDivisionError. The three core fuzz properties fail on minimized inputs including **None** for quota and ratio parsing, and an empty string for tier lookup. The original application is preserved in evidence/before/.

Defensive changes

Validate string types and a 128-character input bound; validate quota digits and units; check the ratio format, finite nonnegative raw size and positive compressed size; reject non-finite results; use safe tier lookup; handle command-line EOF. Invalid function inputs return **None**; the CLI prints an explanatory message and continues.

After the fix

54 tests passed, with no skips or expected failures. Seven Hypothesis properties each completed 300 passing generated examples in the statistics run (2,100 total), supplemented by deterministic regressions, valid examples, length-boundary and CLI checks. This finite test run is evidence of improved robustness, not proof over every possible input.

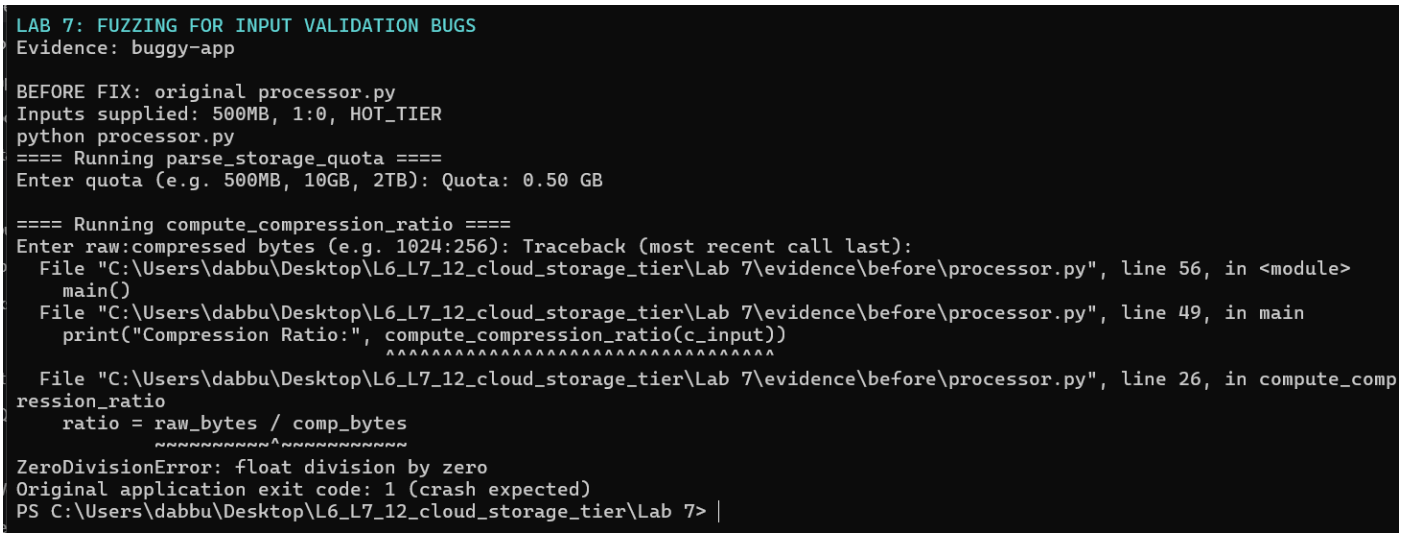
Submission contents

processor.py - fixed application; **test_processor.py** - updated tests; **this PDF** - the four genuine terminal screenshots required by Steps 2, 4, 5 and 6 of the supplied handout. Screenshot timings may differ from saved text logs because they are separate executions.

Step 2 - Run the buggy application

Student-supplied terminal screenshot | Original image embedded without alteration

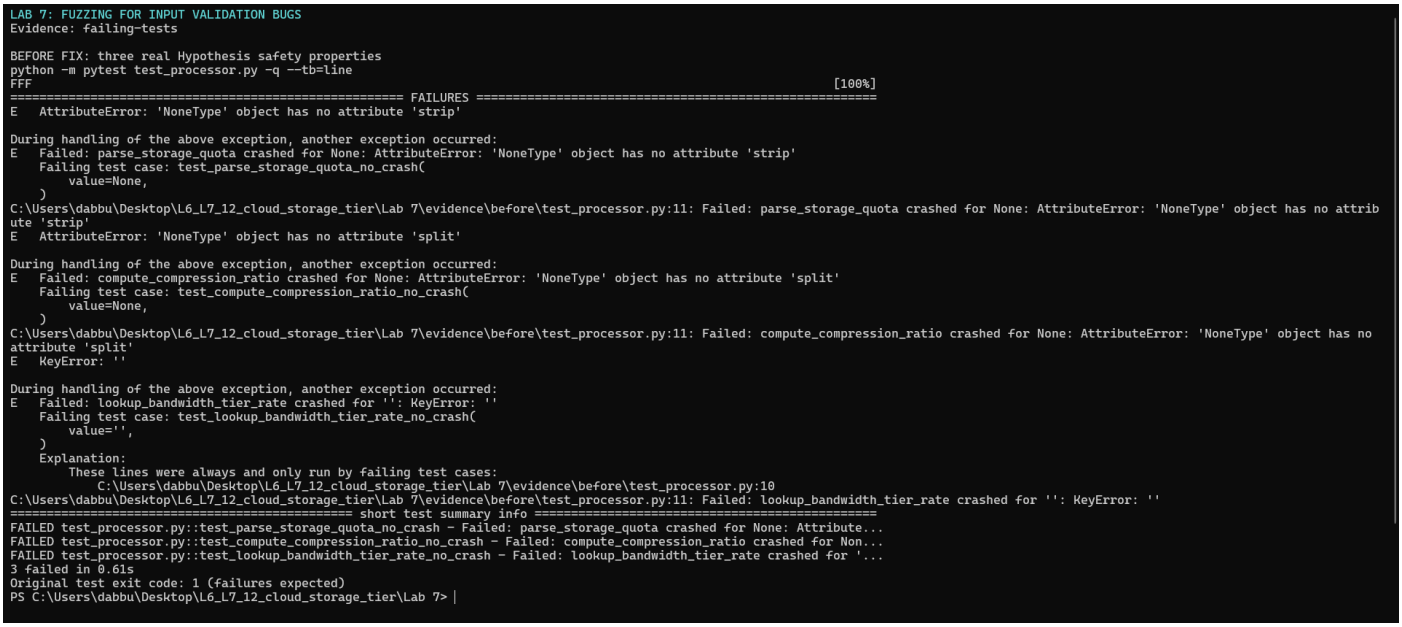
Inputs: 500MB, 1:0, HOT_TIER. The original processor converts the quota, then crashes with ZeroDivisionError while dividing by a zero compressed size. It exits with code 1 before reaching tier lookup.



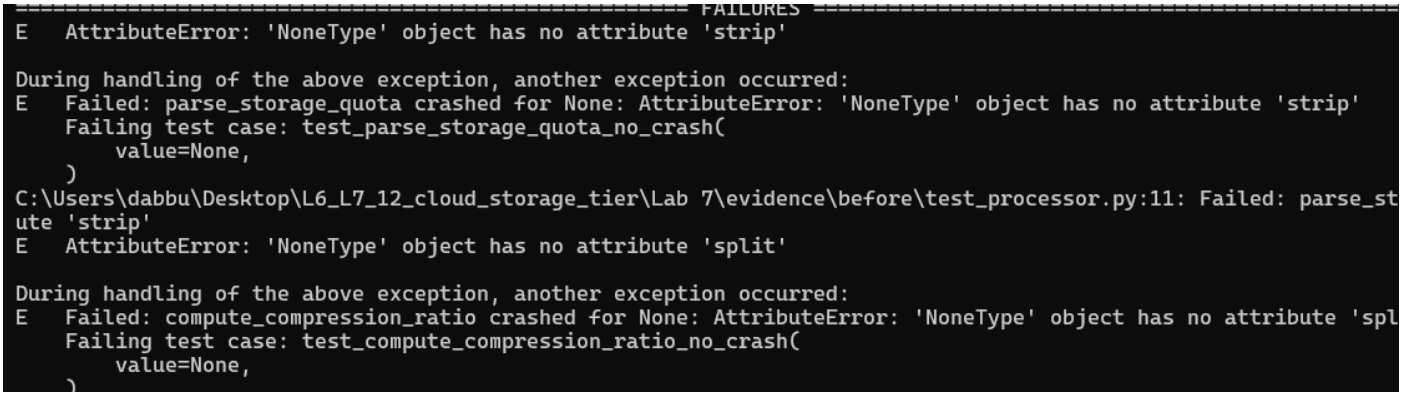
Step 4 - Run the failing fuzz tests

Student-supplied terminal screenshot | Original image embedded without alteration

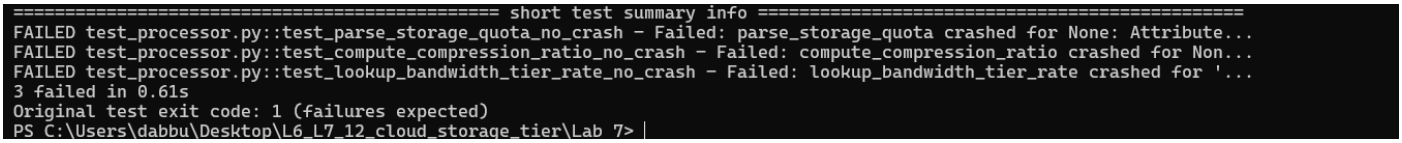
The preserved original fails all three Hypothesis no-crash properties. This screenshot shows the actual minimized inputs and the final 3 failed summary. The tests catch exceptions only to fail explicitly; errors are not swallowed or marked as expected failures.



Enlarged detail: minimized inputs



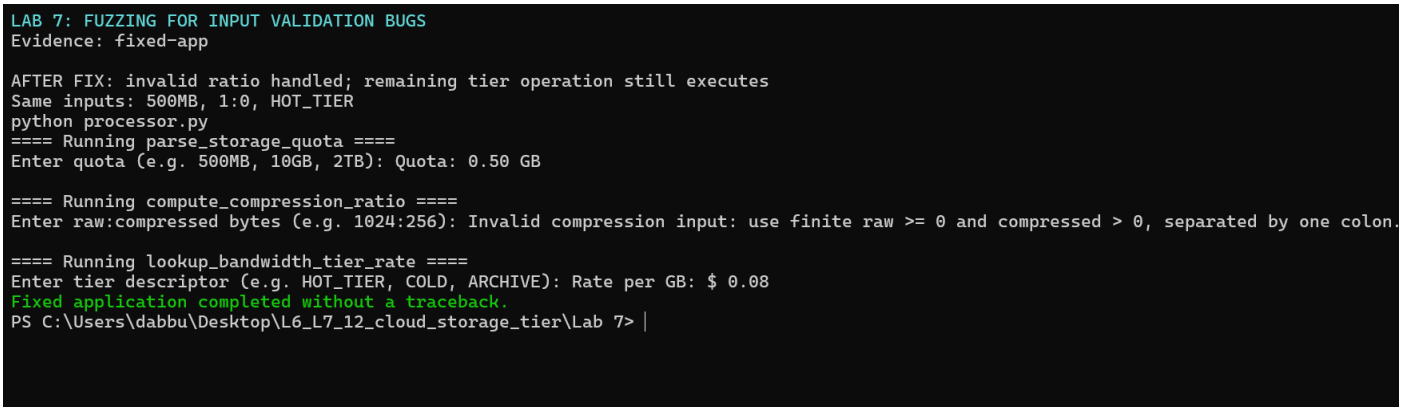
Enlarged detail: failing test summary



Step 5 - Run the fixed application

Student-supplied terminal screenshot | Original image embedded without alteration

The same inputs are used after the fix. The quota succeeds; the zero denominator produces a clear validation message; execution continues to HOT_TIER and returns 0.08. The application exits without a traceback.



Step 6 - Re-run the updated test suite

Student-supplied terminal screenshot | Original image embedded without alteration

All 54 tests pass. The suite includes the same three core safety properties plus valid-input, non-string, regression, boundary and command-line checks. There are no skips or expected failures.

